


# DolphinDos26UL

## User Manual and Documentation

1541 drive ROM and C64 kernal for the Commodore 64 Ultimate (C64U)

Installation and use on the C64U – also runs on DolphinDOS 2 hardware

with JiffyDOS protocol, SD2IEC support and MS-NIBCOPY512

Further development of DolphinDOS 3.0

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# 1 Overview

## 1.1 What is DolphinDos26UL?

DolphinDos26UL is a drive ROM for the 1541 based on the well-known **DolphinDOS 3.0** by Günther Jilg and Jan Bubela. The number **26** stands for the year 2026, **UL** stands for **Ultimate**: this version is built for the emulated drive of the **Commodore 64 Ultimate (C64U)** – parallel cable on the drive's VIA1 (as with DolphinDOS 2.0) and 8 KB of additional drive RAM, without the 512 KB expansion of the Multi-Speeder.

The drive ROM comes with the matching C64 kernal **C64DolphinDos26**. It combines three transfer protocols in a single kernal and, on every access, selects by itself the fastest method that the addressed device supports. On the C64U both files are simply selected in the menu – no soldering, no EPROM programmer.




*Figure 1: Start-up screen of the C64 kernal DolphinDos26 on the C64U*

## 1.2 Which version for which hardware?

There are three drive ROMs with the same DOS core. They differ in the type of parallel cable and in the drive memory:

| Hardware | Drive ROM | Cable and memory | Functions |
|---|---|---|---|
| **Commodore 64 Ultimate (C64U)** | `C1541_DOLPHINDOS26UL.bin` (this manual) | emulated parallel cable on VIA1, 8 KB RAM $6000–$7FFF | DolphinDOS with all @ and X commands, track cache, parallel and block transfer, MS-NIBCOPY512 with REU |
| **Real 1541 with DolphinDOS 2 hardware** | `C1541_DOLPHINDOS26UL.bin` (this manual) | parallel cable on VIA1, 8 KB RAM $6000–$7FFF, 32 KB EPROM | as on the C64U |

| Hardware | Drive ROM | Cable and memory | Functions |
|---|---|---|---|
| **1541 Switchless Multi-Speeder V3** | `1541DolphinDos26.bin` or bank 8 of the Multi-Speeder EPROM (separate manual) | 6821 PIA at $5000, 512 KB drive RAM | additionally **CDR, CDW, CDM** (disk into RAM, RAM disk), MS-Addon-API, MS-MEMTEST512, MS-FASTCOPY512 |
| **Multi-Speeder V3 with parallel cable on VIA1** | `C1541_DOLPHINDOS26DD2.bin` as bank 8 (separate manual) | parallel cable on VIA1, PIA only for the RAM banks, 512 KB drive RAM | as Multi-Speeder V3 |

> **In short:** DolphinDos26UL runs on the C64U and on any real 1541 on which DolphinDOS 2 runs. The variant with **512 KB of drive memory** and the commands **CDR, CDW and CDM** runs **only on the 1541 Switchless Multi-Speeder V3** (chapter 3.2).

### 1.2.1 DolphinDos26UL or DolphinDos26DD2?

Both DOS ROMs transfer in parallel via the drive's VIA1, but they are meant for different hardware. **`C1541_DOLPHINDOS26DD2.bin`** supports the Multi-Speeder's MS-Addons and uses VIA1 for the parallel transfer. **`C1541_DOLPHINDOS26UL.bin`** is specifically compatible with DolphinDOS 2 and needs only the 8 KB track cache and a DolphinDOS 2 / SpeedDOS 2 compatible parallel cable on VIA1. It does not have the MS-Addons such as the internal disk copy and the RAM disk, because they need the 512 KB of the Multi-Speeder V3.

| Feature | C1541_DOLPHINDOS26DD2.bin | C1541_DOLPHINDOS26UL.bin |
|---|---|---|
| Intended for | 1541 with Switchless Multi-Speeder V3 (512 KB) | DolphinDOS 2 hardware (8 KB track cache) and the C64U |
| Size | 64 KB (bank 8 of the EPROM) | 32 KB ($8000–$FFFF) |
| Parallel transfer | VIA1 (SpeedDOS cable) | VIA1 (DolphinDOS 2 / SpeedDOS 2 compatible cable) |
| Drive memory | 512 KB of the Multi-Speeder | only 8 KB track cache at $6000 |
| MS-Addons (internal disk copy CDR/CDW, RAM disk CDM, MS-Addon API) | yes | no |
| MS-FASTCOPY512 / MS-NIBCOPY512 | MS-RAM or REU | REU only |

## 1.3 Files

| File | Content | Checksum (MD5) |
|---|---|---|
| `C1541_DOLPHINDOS26UL.bin` | Drive ROM, 32 KB ($8000–$FFFF): DolphinDOS + CBM-DOS 901229-05 | `e442f8040e0089f10d08edcc6fce353b` |

| File | Content | Checksum (MD5) |
|---|---|---|
| `C1541_DOLPHINDOS26DD2.bin` | 64 KB: DolphinDos26DD2 for bank 8 of the 1541 Switchless Multi-Speeder V3. It runs the Multi-Speeder via VIA1 (SpeedDOS cable) instead of its PIA chip, as DolphinDOS 3 intends | `337b632504ad7ad5131f2e745e3073a8` |
| `C64DolphinDos26.bin` | C64 kernal, 8 KB ($E000–$FFFF) | `a9bf3125240ec59846492a4dd65eaea6` |
| `ms-tools.d64` | MS-Tools disk with MS-NIBCOPY512 (Build 49, BETA, with G64 import and export) , MS-FASTCOPY512 (Build 38, REU memory) and MS-15SECCOPY (Build 16); programs packed with Exomizer | `326fa177db0acac1b717fa670111dd3a` |
| `ms-15seccopy.d64` | Also as a separate disk (in the release in the folder `ms-tools`): MS-15SECCOPY (Build 16); packed with Exomizer | `0d898582c08f3253e85603ed4c2d013a` |

## 1.4 Advantages over DolphinDOS 3

- **Three protocols in one kernal:** DolphinDOS parallel transfer, JiffyDOS and the normal serial Commodore protocol. The original DolphinDOS kernal knew only parallel and serial.
- **SD2IEC and several drives** on one computer: each device is addressed with the best method it supports – without switching.
- **Visible transfer type:** when loading, the kernal shows `LOADING DD:` (DolphinDOS parallel) or `LOADING JF:` (JiffyDOS).
- **SD2IEC function keys** for changing directories and disk images and for loading directly from the directory.
- **Parallel transfer on the C64U:** the C64U emulates only the SpeedDOS-style parallel cable on the drive's VIA1, not the 6821 PIA of the DolphinDOS 3 cable. DolphinDos26UL therefore transfers via the VIA1 – using the parallel routines of DolphinDOS 2.0, including fast loading and saving (XQ/XZ).
- **Adapted for the C64U:** clean cold start of the drive, even if the drive RAM contains random values at power-on (with DolphinDOS 3 this could, for example, leave parallel transfer switched off).
- **Formatting bug fix:** after moving to track 1 the head is always in the correct phase (with DolphinDOS 3 formatting could fail depending on the head position).
- **Faster read and write routines** in the track cache.
- Fully **compatible in operation**: @ commands, X commands, 40 tracks, function keys and editor functions as usual.

# 2 Installation and setup on the C64U

## 2.1 What is needed?

- a **Commodore 64 Ultimate (C64U)** with its emulated drive (drive A, type 1541)
- a USB stick or SD card for the C64U
- the files `C1541_DOLPHINDOS26UL.bin` (drive ROM) and `C64DolphinDos26.bin` (C64 kernal)
- for MS-NIBCOPY512 additionally the disk `ms-tools.d64` and the C64U's REU switched on

On the C64U nothing is soldered and no EPROM is programmed: drive ROM, kernal, drive RAM and parallel cable are set in the C64U menu.

## 2.2 Setup step by step

The exact names of the menu items depend on the firmware version of the C64U; in essence:

1. Copy the files `C1541_DOLPHINDOS26UL.bin` and `C64DolphinDos26.bin` to the USB stick or SD card and insert it into the C64U.
2. Open the C64U menu and set for the emulated **drive A**: drive switched on, type **1541**, device address **8**.
3. As the **1541 ROM** of this drive, choose the file `C1541_DOLPHINDOS26UL.bin` (select the file in the file browser and apply it as the drive ROM). The ROM has 32 KB; according to the source code of the Ultimate firmware, 1541 ROMs of 16 or 32 KB are loaded to $8000–$FFFF.
4. For this drive, switch on the **RAM expansion at $6000–$7FFF** (8 KB, “Extra RAM”). DolphinDOS uses it for its working data and the track cache.
5. Switch on the **parallel cable** for the drive. The C64U emulates the SpeedDOS-style parallel cable on the drive's VIA1 (as with DolphinDOS 2.0) – exactly what DolphinDos26UL is built for.
6. As the **kernal ROM** of the C64, choose the file `C64DolphinDos26.bin` (select it in the file browser and apply it as the kernal).
7. For MS-NIBCOPY512: switch on the **REU** (512 KB or more).
8. Save the settings and **restart** the C64U (cold start, so that kernal and drive ROM are reloaded).

| Setting | Value |
|---|---|
| Drive A | on, type 1541, device 8 |
| 1541 ROM | `C1541_DOLPHINDOS26UL.bin` |
| Drive RAM | 8 KB at $6000–$7FFF |
| Parallel cable | on (VIA1, SpeedDOS style) |
| Kernal ROM | `C64DolphinDos26.bin` |
| REU | on, 512 KB or more (for MS-NIBCOPY512) |

## 2.3 The first start

After power-on the start-up screen appears with colour bars and “DOLPHINDOS26/JIFFY/SD2IEC” (Figure 1). Below it are the note “MULTISPEEDER BY FRANKEGGEN@EMDEN.NET” and “PAL

SYSTEM” or “NTSC SYSTEM”. If the normal Commodore start-up screen appears, the kernal is not loaded.

## 2.4 Checking that it works

Two short entries show whether the drive is set up correctly:

```
@                Fehlerkanal:   73,DOLPHINDOS26 C64U,00,00
@X               X-Status:      02,R+:F+:V+:P+:,00,08
```

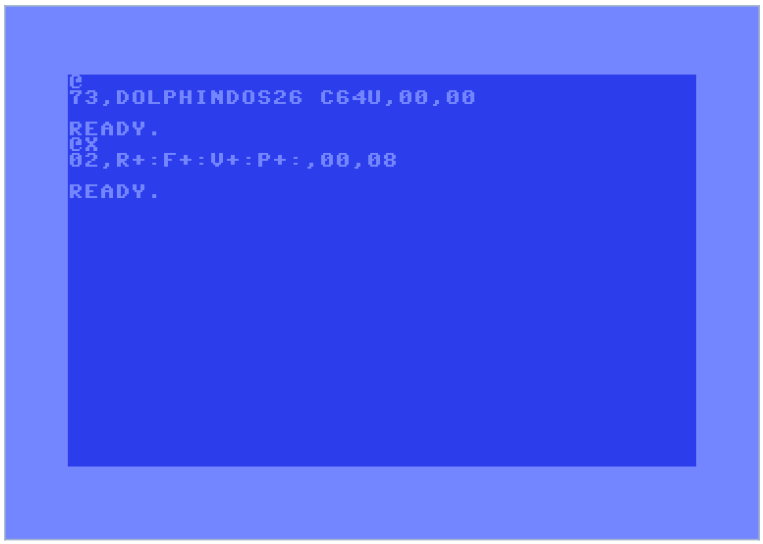



*Figure 2: Error channel with the power-on message of DolphinDos26UL and X status with all switches on*

The power-on message `73,DOLPHINDOS26 C64U` shows that the correct drive ROM is running. Before the first disk access the X status shows `00` as the track count; after the first access the detected track count appears (35 or 40). `P+` shows that parallel transfer is switched on.

## 2.5 Loading the first program

Insert a disk (e.g. `ms-tools.d64`) in drive A in the C64U menu. `@$` shows the directory without erasing a program in memory; then load as usual:

```
@$
LOAD"MS-NIBCOPY512",8
```




*Figure 3: Directory with @$ and loading via the parallel cable – recognisable by "LOADING DD:" with start and end address*

`LOADING DD: $0801-$40C4` means: the program was transferred block by block via the parallel cable. If only `LOADING` without `DD:` appears, the transfer was serial – in that case see the troubleshooting in the next section.

> **Important:** The C64U does not emulate a 6821 PIA at $5000 (DolphinDOS 3 cable), only the parallel cable on the drive's VIA1. DolphinDos26UL therefore transfers in parallel via VIA1 port A ($1801) with the handshake lines CB1/CA2 – using the parallel routines of DolphinDOS 2.0. The C64 side (kernal `C64DolphinDos26.bin`) is the same for both cables.

## 2.6 Troubleshooting

| Observation | Cause and remedy |
|---|---|
| Normal Commodore start-up screen instead of "DOLPHINDOS26/JIFFY/SD2IEC" | Kernal not loaded: choose `C64DolphinDos26.bin` as the kernal ROM, save the settings, cold start. |
| `@` shows `73,CBM DOS V2.6 1541` or another message | Drive A is not running DolphinDos26UL: choose `C1541_DOLPHINDOS26UL.bin` as the 1541 ROM. |
| Drive does not respond or hangs | The drive's RAM expansion $6000–$7FFF is not switched on – DolphinDOS needs it for its working data. |
| `LOADING` without `DD:` | Parallel cable not switched on in the C64U menu or switched off with `@XP-` (`@X` then shows `P-`); switch on again with `@XP+`. |
| Loading hangs at `LOADING` | Wrong ROM for the C64U: original DolphinDOS 3 and `1541DolphinDos26.bin` (Multi-Speeder) transfer via the PIA, which the C64U does not have. Use only `C1541_DOLPHINDOS26UL.bin`. |
| MS-NIBCOPY512 shows `NEED REU` | Switch on the REU in the C64U menu and restart MS-NIBCOPY512. |
| Saved game has only 0–1 blocks or loading has no effect (e.g. Eye of the Beholder) | Older C64 kernal: use the current `C64DolphinDos26.bin` (as of 5 November 2026) – it saves and loads saved games of programs with their own interrupt completely. |
| A game or program does not run | Step-by-step remedy in chapter 6.11. |

## 2.7 Back to the original configuration

In the C64U menu, choose the built-in 1541 ROM for drive A again and the built-in kernal for the C64, switch off the RAM expansion and parallel cable if required, and restart. Disks and files are not changed by DolphinDos26UL – disks saved with DolphinDOS are normal 1541 disks (those formatted with 40 tracks only in drives that read 40 tracks).

# 3 Operation on real hardware

## 3.1 Real 1541 with DolphinDOS 2 hardware

DolphinDos26UL also runs on **real hardware on which DolphinDOS 2 runs**. The drive side of DolphinDOS 2.0 has exactly the setup that the C64U emulates: a 32 KB ROM at $8000–$FFFF, 8 KB RAM at $6000–$7FFF and the parallel cable on the VIA1. The parallel routines of DolphinDos26UL come from DolphinDOS 2.0.

| Part | DolphinDOS 2 setup | with DolphinDos26UL |
|---|---|---|
| Drive ROM | 32 KB EPROM with DolphinDOS 2.0 | Program `C1541_DOLPHINDOS26UL.bin` into a 27C256 (32 KB) and replace the DolphinDOS 2 EPROM |
| Drive RAM | 8 KB at $6000–$7FFF | unchanged |
| Parallel cable | VIA1 port A (PA0–PA7) of the drive to the C64 user port, handshake via CB1/CA2 | unchanged |
| C64 kernal | DolphinDOS kernal | Program `C64DolphinDos26.bin` into an 8 KB EPROM (2764/27C64) or load it into the kernal switcher |

Operation and commands are the same as on the C64U; the power-on message is also `73,DOLPHINDOS26 C64U`. MS-NIBCOPY512 detects the VIA1 cable by itself and needs a REU in the C64.

> **Note:** A 1541 with the DolphinDOS 3.0 cable (6821 PIA at $5000) is not a DolphinDOS 2 setup – DolphinDos26UL does not transfer in parallel there. The Multi-Speeder version `1541DolphinDos26.bin` is built for the PIA.

## 3.2 1541 Switchless Multi-Speeder V3 (512 KB)

The variant with **512 KB of drive memory** is called **DolphinDos26** (`1541DolphinDos26.bin`, the last bank in the Multi-Speeder EPROM) and runs **only on the 1541 Switchless Multi-Speeder V3**. This board provides the 6821 PIA at $5000, through which the parallel cable runs and which at the same time maps one of 64 RAM banks of 8 KB each into the drive. Only with it are these commands possible:

| Command | Effect (Multi-Speeder V3 only) |
|---|---|
| `@CDR1` / `@CDR2` | read a whole disk (35 tracks) into side 1 or 2 of the drive RAM – without transfer to the C64; `@CDR1+`/`@CDR2+` with 40 tracks |
| `@CDW1` / `@CDW2` | write side 1 or 2 to the inserted disk – any number of copies in succession |
| `@CDM1` / `@CDM2` / `@CDM0` | RAM disk: drive 8 works directly from side 1 or 2 (LOAD, SAVE, directory without motor and head); `@CDM0` switches back to the disk |
| MS-Addon-API | Jump table for C64 programs; used by MS-MEMTEST512 and MS-FASTCOPY512 (MS-RAM) |

In DolphinDos26UL, CDR, CDW and CDM report `31,SYNTAX ERROR` because the C64U and the DolphinDOS 2 hardware have only 8 KB of drive RAM. Everything else about the Multi-Speeder is in its separate manual `DolphinDos26_Handbuch.odt`, which also describes `C1541_DOLPHINDOS26DD2.bin`, the Multi-Speeder version with the parallel cable on VIA1.

| Property | DolphinDos26UL | DolphinDos26 (Multi-Speeder V3) |
|---|---|---|
| Hardware | C64U, real 1541 with DolphinDOS 2 hardware | 1541 Switchless Multi-Speeder V3 only |
| Parallel cable | VIA1 ($1801, CB1/CA2) | 6821 PIA ($5000, CA1/CA2) |
| Drive RAM | 8 KB | 512 KB in 64 banks |
| DolphinDOS commands, track cache, 40 tracks | yes | yes |
| CDR / CDW / CDM | – (`31,SYNTAX ERROR`) | yes |
| MS-NIBCOPY512 | yes, storage in the REU | yes, storage in the REU or in the Multi-Speeder RAM |
| MS-FASTCOPY512 | yes, memory in the REU | yes, memory in the Multi-Speeder RAM or the REU |
| MS-MEMTEST512 | – | yes |

# 4 Transfer protocols and multiple drives

## 4.1 The three transfer protocols

On every device access the kernal checks which method is possible and uses it in this order:

| Rank | Protocol | When? | Display when loading |
|---|---|---|---|
| 1 | **DolphinDos26 parallel** | The drive has DolphinDOS and is connected via the parallel cable. This is detected at every secondary address by a response pulse from the floppy. | `LOADING DD: $0801-$xxxx` |
| 2 | **JiffyDOS** | The device identifies itself as a JiffyDOS device during TALK/LISTEN (e.g. SD2IEC, 1541 with JiffyDOS ROM). All bytes then run in the Jiffy protocol: LOAD, SAVE, SEQ files, directory, status. | `LOADING JF: $0801-$xxxx` |
| 3 | **Standard DOS (serial)** | All other devices, e.g. an original 1541, 1571, 1581 or a printer. | `LOADING` |

When loading with secondary address 2 (`LOAD"name",8,2`, compatible mode) and for the directory (`LOAD"$",8`), transfer is not block by block. The bytes nevertheless run via the parallel cable or in the Jiffy protocol; the display is then `LOADING DD:` or `LOADING JF:` without addresses. In addition, Jiffy fast loading is not used for load addresses below $0400.

> **Note:** From the JiffyDOS system, the **transfer protocol** has been adopted, not the command and key interface of the JiffyDOS kernal. The computer is operated as with DolphinDOS.

### 4.1.1 Speed

Measured in the emulator VICE with DolphinDos26 (same DOS core as DolphinDos26UL), program with 16 000 bytes:

| Operation | DolphinDos26 parallel | JiffyDOS (1541) | Standard DOS |
|---|---|---|---|
| LOAD | 2.1 s | 6.7 s | 41.2 s |
| LOAD …,8,2 | – | 15.8 s | – |
| SAVE | 4.6 s | 19.1 s | 48.4 s |

## 4.2 Multiple drives and SD2IEC devices

Because the selection is made individually for each device, different drives can be operated together. A typical setup on the C64U:

| Device | Example | Protocol |
|---|---|---|
| 8 | Drive A of the C64U with DolphinDos26UL, parallel cable active | DolphinDos26 parallel |
| 9 | second drive with JiffyDOS ROM | JiffyDOS |
| 10 | SD2IEC on the serial port | JiffyDOS |
| 11 | original 1541/1571/1581 | Standard DOS |

- Only **the one drive** that is connected to the parallel cable and has DolphinDOS can ever work in parallel.
- `@9` (in general `@` + digit) sets the **default device** on the C64 for LOAD, SAVE and the @ commands; `@8` switches back.
- The **device address of a DolphinDOS drive** is changed with `@X9",8` (see chapter 6.7).
- Dedicated function keys are available for SD2IEC devices (chapter 6.9).

# 5 New features and improvements

## 5.1 The drive ROM DolphinDos26UL

| New feature | Description |
|---|---|
| 32 KB ROM for the C64U | The ROM occupies $8000–$FFFF: extension code from $8000, DolphinDOS at $A000–$BFFF, the adapted CBM-DOS 901229-05 at $C000–$FFFF. |
| Power-on message | `73, DOLPHINDOS26 C64U` (instead of `73, DOLPHINDOS25 1541`). |
| Parallel cable on VIA1 | Transfer via VIA1 port A ($1801) with the handshake lines CB1/CA2, as emulated by the C64U (routines from DolphinDOS 2.0): cable detection, single bytes, block transfer for fast loading and saving. The C64 kernal is the same for both cable types; when loading a file with 119 blocks, the VIA path is as fast as the PIA path in the emulator. |
| Clean cold start | DolphinDOS assumes that the drive RAM is cleared at power-on. On the C64U it contains random values. DolphinDos26UL detects this by a missing signature (bytes $A5 $5A at $60F0), then clears $6000–$60FF, switches on all switches R/F/V/P, sets automatic track detection and sets the signature. A later reset of the drive (`@UJ`) keeps the settings. |
| Formatting repaired | After moving to track 1 (bump), the phase of the stepper motor now always matches track 1. Previously, depending on the previous head position, formatting took place one track too far in, and formatting could abort with `24,READ ERROR,19,00`. |
| Faster reading | Unrolled GCR decoding in the track cache: one sector 12.0 → 10.4 ms. |
| Faster writing | Optimised copy of the write buffer: one sector 11.8 → 10.5 ms. |
| Clear error message | Commands known only to the Multi-Speeder (CDR, CDW, CDM) and unknown X commands report `31,SYNTAX ERROR`. |

## 5.2 The C64 kernal DolphinDos26

| New feature | Description |
|---|---|
| JiffyDOS protocol | Full protocol (LOAD with fast loading, SAVE, all bytes) for JiffyDOS devices; the routines are identical instruction for instruction to the JiffyDOS kernal. DolphinDOS parallel takes priority. |
| LOADING DD: / JF: | `LOADING` and `VERIFYING` show the transfer type: `DD:` for DolphinDOS parallel, `JF:` for JiffyDOS, no marker for serial standard DOS. |
| SD2IEC function keys | C=F3 one directory/image back, C=F5 change to the directory/image of the directory line, C=F7 load the file of the directory line. |
| Colours as in the original | Border $0E, background $06, text $0E as in the original C64 ROM; the texts of the start-up screen also appear in the text colour. To change them yourself, see “Personalisation”. |
| New start-up screen | Colour bars, “DOLPHINDOS26/JIFFY/SD2IEC”, the note “MULTISPEEDER BY FRANKEGGEN@EMDEN.NET” and “PAL SYSTEM” or “NTSC SYSTEM” (Figure 1). |
| PAL/NTSC detection | Via the raster line; the keyboard IRQ is set accordingly. |
| F2 = RUN: | F2 writes `RUN:` (previously `SYS$0`). |

| New feature | Description |
|---|---|
| Saved games | Parallel LOAD and SAVE check the STOP key via the kernal routine STOP (abort only when the key is pressed). The original kernal evaluated only one bit and aborted if a program with its own interrupt did not maintain the keyboard status: as a result, Eye of the Beholder saved only an empty file with 0–1 blocks and did not load saved games. Now the saved game is saved completely (88 blocks) and loaded again. |
| More room in the ROM | Monitor, Centronics driver, cassette and RS-232 remnants removed; 153 bytes free for future extensions. All central entry points of the kernal are unchanged. |

## 5.3 Personalisation

The colours are single bytes in the ROM and can be changed without an assembler: copy the file `C64DolphinDos26.bin` (8 KB, address $E000 = file offset $0000), change the bytes with a hex editor and use the copy as the kernal. The C64 does not check a kernal checksum. The factory values are those of the original C64 ROM.

| Colour | ROM address | File offset | Factory value |
|---|---|---|---|
| Border | $ECD9 | $0CD9 | $0E (light blue) |
| Background | $ECDA | $0CDA | $06 (blue) |
| Text (also start-up screen, `READY.`, cursor, CTRL-V) | $E535 | $0535 | $0E (light blue) |

Colour values for these three bytes:

| Value | Colour | Value | Colour |
|---|---|---|---|
| $00 | black | $08 | orange |
| $01 | white | $09 | brown |
| $02 | red | $0A | light red |
| $03 | cyan | $0B | grey 1 |
| $04 | purple | $0C | grey 2 |
| $05 | green | $0D | light green |
| $06 | blue | $0E | light blue |
| $07 | yellow | $0F | grey 3 |

All texts of the start-up screen follow the text colour. Only the colour bars have their own colours, as PETSCII control codes in the text:

| Bar | ROM address | File offset | Factory code |
|---|---|---|---|
| 1 | $F001 | $1001 | $1C (red) |
| 2 | $F02D | $102D | $81 (orange) |
| 3 | $F03F | $103F | $9E (yellow) |
| 4 | $F06B | $106B | $1E (green) |
| 5 | $F079 | $1079 | $9A (light blue) |
| 6 | $E473 | $0473 | $9C (purple) |

PETSCII control codes of the colours:

| Code | Colour | Code | Colour |
|---|---|---|---|
| $90 | black | $81 | orange |
| $05 | white | $95 | brown |
| $1C | red | $96 | light red |
| $9F | cyan | $97 | grey 1 |
| $9C | purple | $98 | grey 2 |
| $1E | green | $99 | light green |
| $1F | blue | $9A | light blue |
| $9E | yellow | $9B | grey 3 |

- Do not choose the same text and background colour, or nothing will be readable.
- The control code $06 (`PRINT CHR$(6);`) is new and selects the text colour from $E535. The start-up screen uses it after each bar.
- The addresses apply to this version of the kernal; with later versions compare the existing byte before changing it.
- A changed file has a different checksum from the published one.

# 6 Using the DOS commands

## 6.1 Quick reference

| Input | Effect |
|---|---|
| `@` | Read the error channel (status) of the drive |
| `@befehl` | Send a command to the drive, e.g. `@S:DATEI` |
| `@$` / `@$:A*` | Show the directory without erasing the program in memory (also with a pattern) |
| `@8` … `@9` | Set the default device on the C64 |
| `LOAD"name",8` / `,8,1` | Load (BASIC start / to the address from the file) |
| `LOAD"name",8,2` | compatible loading (not block by block) |
| `LOAD` / `VERIFY` | without a name: first file (`:*`) or last used file (`*`) |
| `SAVE"@:name",8` | Replace a file (the old file is deleted first) |
| `@N:name,id` / `@N:name,id+` | Format a disk, 35 / 40 tracks |
| `@S:name` | Delete file(s) |
| `@R:neu=alt` | Rename a file |
| `@C:neu=alt` | Copy a file |
| `@V` | Tidy up the disk (Validate) |
| `@I` / `@UJ` | Initialise the disk / reset the drive |
| `@X` | X status: `02,R+:F+:V+:P+:,35,08` |
| `@XR±` `@XF±` `@XV±` `@XP±` | Track cache / fast LOAD-SAVE / verify / parallel cable on-off |
| `@XT+` `@XT-` `@XT` | 40 tracks fixed / 35 tracks fixed / automatic |
| `@X+` / `@X-` | Default settings / largely like CBM-DOS |
| `@XL:name` / `@XU:name` | Lock / unlock a file |
| `@X9",8` | Change the device address of drive 8 to 9 |
| `@CD:verz` `@CD←` | SD2IEC: change directory/image, back |
| `&` `&"name"` `&*` `&zahl` | Reset / load F keys, OLD, convert number |
| `SYS $C000` | SYS with hexadecimal address |
| F1–F8, C=F1–C=F7 | Function keys (chapter 6.10) |
| SHIFT+RUN/STOP | Load and start the first file |

## 6.2 Sending commands

In **direct mode**, commands are most easily entered with the at sign `@`: `@` at the start of the line, then the command, finished with RETURN. The kernal sends the text unchanged via command channel 15 and then shows the error channel. Quotation marks are not needed.

```
@S:ALTDATEI
01, FILES SCRATCHED,01,00
```

In **programs** and with another kernal, the same command works via the command channel:

```
10 OPEN 1,8,15,"S:ALTDATEI": CLOSE 1
20 OPEN 1,8,15: INPUT#1,A,B$,C,D: PRINT A;B$;C;D: CLOSE 1
```

The @ commands go to the **default device** (default 8). `@9` makes device 9 the default device, `@8` sets it back. The default device also applies to LOAD and SAVE without a device address and to CTRL-D.

## 6.3 Status and error channel

`@` alone reads the error channel. The message has the form *number, text, track, sector*. Important messages:

| Message | Meaning |
|---|---|
| `00, OK,00,00` | no error |
| `01, FILES SCRATCHED,nn,00` | nn files deleted |
| `02,R+:F+:V+:P+:,35,08` | X status (see chapter 6.7) |
| `20–29` | Read error (sector not found, checksum, write protection at 26 …) |
| `31,SYNTAX ERROR` | unknown command; also for CDR/CDW/CDM (Multi-Speeder only) |
| `33,SYNTAX ERROR` | wildcard (`*`, `?`) in the name when writing |
| `62,FILE NOT FOUND` | file not found |
| `63,FILE EXISTS` | file exists; also when replacing a locked file with `SAVE"@:…"` |
| `72,DISK FULL` | disk full |
| `73, DOLPHINDOS26 C64U` | power-on message |
| `74,DRIVE NOT READY` | no disk |

## 6.4 Directory

`@$` shows the directory directly on the screen without overwriting the BASIC program in memory. A pattern restricts the display, e.g. `@$:SPIEL*`. RUN/STOP aborts the display. Alternatively: CTRL-D or F7. The classic `LOAD"$",8` and `LIST` also work.

## 6.5 Loading and saving

| Input | Effect |
|---|---|
| `LOAD"name",8` | Load to the BASIC start |
| `LOAD"name",8,1` | Load to the address from the file |
| `LOAD"name",8,2` | compatible loading (byte by byte); helps with programs that have problems with fast loading |
| `LOAD` | without a name: first file on the disk (`:*`) |
| `VERIFY` | without a name: last used file (`*`) |
| `SAVE"name",8` | Save |
| `SAVE"@:name",8` | Replace: DolphinDOS deletes the old file first and then saves anew. Locked files are not replaced (63,FILE EXISTS). |
| SHIFT+RUN/STOP | Load and start the first file; on a directory line, this file |

In direct mode the end address is shown after loading, e.g. `LOADING DD: $0801-$0BAA`. When loading to the BASIC start, the kernal resets the program end; machine language programs loaded elsewhere thus do not destroy a BASIC program.

## 6.6 Disk commands

| Command | Effect |
|---|---|
| `@N:name,id` | format with 35 tracks (664 blocks free) |
| `@N:name,id+` | format with 40 tracks (749 blocks free) |
| `@N:name` | erase only the directory (fast, disk must be formatted) |
| `@S:name` | delete; wildcards allowed (`@S:TEST*`), several names with a comma |
| `@R:neu=alt` | rename |
| `@C:neu=alt` | copy a file; `@C:neu=a,b` concatenates files |
| `@V` | Validate: rebuild the allocation table (BAM); especially fast with the track cache switched on |
| `@I` | initialise the disk |
| `@UJ` | reset the drive (the DolphinDOS settings are retained) |
| `M-R`, `M-W`, `M-E`, `B-R`, `B-W`, `U1`, `U2` … | memory and block commands as in CBM-DOS (channel 15) |

DolphinDOS formats quickly: the head moves only as far as necessary, the track gap is measured only at the start of each density zone, and verify reading takes place only when verify (`XV+`) is switched on.

## 6.7 X commands and X status

`@X` shows the X status, e.g.:

```
02,R+:F+:V+:P+:,35,08
```

| Part | Meaning |
|---|---|
| `R+` / `R-` | Track cache: a whole track is held in the 8 KB RAM |
| `F+` / `F-` | fast block-by-block LOAD/SAVE; `F-` works similarly to `,8,2` |
| `V+` / `V-` | verify reading (Verify) after writing |
| `P+` / `P-` | parallel transfer via the parallel cable |
| `!` | after the last colon: track count fixed (`XT+`/`XT-`) |
| `35` / `40` | track count of the disk; `00` before the first access or after a disk change |
| `08` | device address |

| Command | Effect |
|---|---|
| `@XR+` / `@XR-` | track cache on / off |
| `@XF+` / `@XF-` | fast LOAD/SAVE on / off |
| `@XV+` / `@XV-` | verify on / off (off: writing and formatting faster, but without checking) |
| `@XP+` / `@XP-` | parallel transfer on / off |

| Command | Effect |
|---|---|
| `@XT+` / `@XT-` / `@XT` | 40 tracks fixed / 35 tracks fixed / automatic detection on first access |
| `@X+` | default settings: all switches on (closes open files) |
| `@X-` | track cache and parallel off, 35 tracks fixed – largely like CBM-DOS |
| `@XL:name` / `@XU:name` | lock / unlock a file (wildcards allowed); locked files carry a `<` in the directory |
| `@X9",8` | change the device address (4–14); the quotation mark prevents waiting for the status under the old address |

The settings are retained on a reset of the drive (`@UJ`). Only a cold start (switching off) restores the default settings.

## 6.8 Multiple drives

- `@9` makes device 9 the default device, `@8` sets it back. LOAD/SAVE with a device address (`,9`) work independently of this.
- With `@X9",8` a DolphinDOS drive is given address 9 permanently until it is switched off.
- The transfer type is determined individually for each device (chapter 4.1); `LOADING DD:`/`JF:` shows which one was used.

## 6.9 SD2IEC devices

SD2IEC devices are addressed via the JiffyDOS protocol. Their commands (e.g. for changing directories and disk images) are sent like DOS commands with `@`; details are in the SD2IEC manual.

| Input / key | Effect |
|---|---|
| `@CD:name` | change to the directory or disk image (.D64 …) |
| `@CD←` or **C=F3** | one directory or disk image back |
| `@CD//` | to the root directory |
| **C=F5** on a directory line | change to this directory/image (`"CD:` before the name) |
| **C=F7** on a directory line | writes `LOAD"name":` – load with RETURN, from the device set with `@9`, otherwise device 8, to the address in the file |

For C=F5 and C=F7, first show the directory with `@$`, then move the cursor to the desired line. Directory lines with a five-digit block count (SD2IEC) are automatically aligned correctly.

### 6.9.1 SD2IEC on the C64U

The C64U itself also offers SD2IEC, with no additional device. In the menu under `SoftIEC Drive Settings`, `IEC Drive` turns the drive on, `Soft Drive Bus ID` sets the device number (one that no other drive uses, e.g. 9) and `Default Path` the start directory, e.g. `/USB1/SD2IEC/`. The drive speaks the JiffyDOS protocol (`LOADING JF:`). `@9` makes it the default device until the next reset.




*Figure 4: SD2IEC on the C64U: changing directory and inserting a disk image*




*Figure 5: C=F7 on a directory line, then RETURN: loading via JiffyDOS*

The drive reports itself as `73,U64IEC ULTIMATE DOS V1.1`. It is a separate reproduction of SD2IEC and does not know all commands. With C64U firmware 1.1.0 the following applies:

| Command | Effect on the C64U |
|---|---|
| `@CD:GAMES` | change into the subdirectory GAMES |
| `@CD:GAMES/ARCADE` | several levels at once |
| `@CD:..` | one level back |
| `@CD←` or **C=F3** | back to the start directory (`Default Path`), not just one level |
| `@CD//` | to the top level of the C64U |
| `@CD:/USB1/SD2IEC` | full path |
| `@CD:GAME.D64` | insert a disk image; leave with `@CD←` |
| `@MD:NEW`, `@RD:NEW` | create and delete a directory, only in the current directory |
| `@S:NAME`, `@R:NEW=OLD`, `@C:COPY=ORIGINAL`, `@I` | scratch, rename, duplicate a file; initialise the drive |

Not available are the forms `CD/name` and `MD/…`, the X commands, and `T-R`, `G-P`, `CP`, `UJ`, `V`, `N` and `L`. The drive remembers its directory across a reset of the C64. Measured with 16 KB: LOAD 1.8 s, SAVE 5.5 s.

## 6.10 Keyboard and screen

### 6.10.1 Function keys

| Key | Text | Key | Text |
|---|---|---|---|
| F1 | CLR, `LIST` | C=F1 | `OPEN4,4:CMD4:LIST` (listing to printer) |
| F2 | `RUN:` | C=F3 | `@CD←` (SD2IEC back) |
| F3 | CLR, `RUN` | C=F5 | `"CD:` (SD2IEC change) |
| F4 | `VERIFY` | C=F7 | `LOAD"name":` |
| F5 | `LOAD` | | |
| F6 | `SAVE"@:` | | |
| F7 | CLR, `@$` | | |
| F8 | `@X` | | |

The function keys work outside quote and insert mode. CTRL-X switches them off (together with the additional functions), clearing $0293–$0299 and $0334–$03FF in the process; `&` restores the standard assignment. A custom assignment is loaded from disk with `&"name"`.

### 6.10.2 CTRL functions

| Keys | Effect |
|---|---|
| CTRL-D | Show the directory of the default device |
| CTRL-@ | Show the floppy status |
| CTRL-A | Key repeat for all keys on/off |
| CTRL-B | Cursor to the start of the last screen line |
| CTRL-G | Cursor 20 columns to the right (tab) |
| CTRL-K | Delete the line from the cursor to the end |
| CTRL-L | Delete the line from the start to the cursor |
| CTRL-V | Restore standard colours, text mode and input/output |
| CTRL-X | Function keys and additional functions off (on again with `&`) |
| CTRL-DEL / C=DEL | Delete a character into a buffer / insert it again |
| CTRL-* | Screen hardcopy to the printer (device 4) |
| CTRL-RETURN | Halt the program until CTRL is released |
| Hold CTRL | Pause scrolling while CTRL is pressed |
| C= | Freeze scrolling until CTRL is pressed |

### 6.10.3 BASIC extensions

| Input | Effect |
|---|---|
| `&` | Reset the function keys to the standard assignment |

| Input | Effect |
|---|---|
| `&"name"` | Load a function key assignment from disk |
| `&*` | OLD: restore a BASIC program after NEW |
| `&49152` / `&$C000` | Output a number in hexadecimal and decimal: `$C000=49152` |
| `SYS $C000` | SYS with hexadecimal address |

### 6.10.4 Reset and RESTORE with an additional key

If one of the following keys is held during reset or when pressing RESTORE, the kernal performs a special action:

| Key | Effect |
|---|---|
| RUN/STOP | BASIC warm start with reset of the vectors (like RUN/STOP-RESTORE) |
| Q | BASIC warm start, vectors are retained |
| SPACE | Cold start without checking for a cartridge at $8000 |
| ← and 1 | Clear the entire memory, then cold start |

## 6.11 If a program does not run

Programs with their own fast loader or copy protection do not always get along with DolphinDOS. The proven sequence of steps, reloading each time:

1. `LOAD"name",8,2` (compatible loading)
2. `@XF-` (fast LOAD/SAVE off)
3. CTRL-X (function keys and additional functions off)
4. `@XR-` (track cache off)
5. `@XP-` (parallel transfer off) or `@X-`
6. switch to the original 1541 ROM and the original kernal in the C64U menu

Titles with “NTSC” or “USA” in the name usually run only on an NTSC system.

# 7 Changes and fixes compared with DolphinDOS 3

## 7.1 Drive

| Area | DolphinDOS 3 | DolphinDos26UL |
|---|---|---|
| Formatting | After moving to track 1 the stepper motor could be in a wrong phase; the disk was then formatted offset, abort with 24,READ ERROR possible | Phase is set correctly, formatting always correct |
| Parallel cable | 6821 PIA at $5000 (port A, CA1/CA2) | VIA1 port A $1801 (CB1/CA2), SpeedDOS-style cable as on the C64U; routines from DolphinDOS 2.0 |
| Cold start | assumes cleared drive RAM; with random content (C64U) e.g. parallel off | detects the cold start by a signature and initialises $6000–$60FF |
| Reading/writing in the track cache | Original routines | faster GCR decoding and buffer copy |
| ROM | 16 KB DolphinDOS ROM | 32 KB for the C64U ($8000–$FFFF) |
| Power-on message | `73, DOLPHINDOS …` | `73, DOLPHINDOS26 C64U` |
| Unknown commands | – | Multi-Speeder commands and unknown X commands: `31,SYNTAX ERROR` |

All known DolphinDOS features have been adopted unchanged: track cache, parallel and block transfer, all X commands, 35/40 tracks with automatic detection, fast formatting, fast deleting and Validate, locking of files and safe replacing with `SAVE"@:…"`.

## 7.2 C64 kernal

| Area | DolphinDOS 3 kernal | DolphinDos26 kernal |
|---|---|---|
| Protocols | DolphinDOS parallel, serial | additionally full JiffyDOS protocol |
| LOAD display | `LOADING` | `LOADING DD:` / `LOADING JF:` / `LOADING` |
| SD2IEC | – | Function keys C=F3, C=F5, C=F7 |
| F2 | `SYS$0` | `RUN:` |
| C=F3 … C=F7 | Monitor functions | SD2IEC functions |
| Machine language monitor | built in (also CTRL + Reset/RESTORE) | **removed**; CTRL + Reset/RESTORE without function; `SYS 0` and BRK lead to a BASIC warm start |
| Centronics printer on the user port | built in (device 4) | **removed**; device 4 is always a serial printer, the hardcopy (CTRL-*) goes there |
| Datasette | Polling of the datasette key in the IRQ | removed (cassette was already not supported in DolphinDOS) |
| RS-232 | Status remnants in READST | removed; READST always returns the status of the serial bus |
| UNLISTEN | CLK remained set | CLK is released as in the original kernal |

| Area | DolphinDOS 3 kernal | DolphinDos26 kernal |
|---|---|---|
| STOP during parallel LOAD/SAVE | only bit 7 of the keyboard status checked: abort in programs with their own interrupt (e.g. Eye of the Beholder: saved game 0–1 blocks, loading without effect) | Kernal routine STOP, abort only when the key is pressed; saved games are saved and loaded completely |
| Colours | border and background black, text cyan | as in the original C64 ROM: border $0E, background $06, text $0E |
| Start-up screen | DolphinDOS message | Colour bars, “DOLPHINDOS26/JIFFY/SD2IEC”, note, PAL/NTSC |

## 7.3 Not included in DolphinDos26UL

The following functions exist only in the Multi-Speeder version `1541DolphinDos26.bin`, because they need the 512 KB RAM banks. In DolphinDos26UL they report `31,SYNTAX ERROR`:

- `CDR1`, `CDR2`, `CDR1+`, `CDR2+` – read a disk into the drive RAM
- `CDW1`, `CDW2` – write RAM to disk
- `CDM1`, `CDM2`, `CDM0` – RAM disk
- MS-Addon-API (jump table from $3FFF) – hence MS-MEMTEST512 does not run, and MS-FASTCOPY512 only with a REU

# 8 MS-NIBCOPY512

## 8.1 Overview

MS-NIBCOPY512 is a nibble copier modelled on **nibtools**, ported to the C64 and with a convenient user interface. A small program (payload, 1279 bytes) is loaded into the drive's RAM and reads each track bit-exactly as raw data – including syncs, gaps, non-standard formats, tracks 36–42 and half tracks. The evaluation (track length, alignment, density, sector check, weak bits) is done by the C64. In this way many copy-protected disks can also be copied.

**BETA:** MS-NIBCOPY512 is in BETA and still under development. Functions and operation may still change, and it has only been partly tested on real hardware. The user interface therefore shows `MS-NIBCOPY512 BETA`.

**Without the Switchless Multi-Speeder board** (i.e. with DolphinDos26UL on the C64U), MS-NIBCOPY512 works **only via the parallel cable**, with the **8 KB of drive RAM** (track cache at $6000–$7FFF) as the track buffer and with **REU memory** in the C64 as the storage for the tracks. Storage in the Multi-Speeder RAM (MS-RAM) is possible only with the Multi-Speeder board (512 KB).

**Cable:** On start, the drive part (payload) of MS-NIBCOPY512 detects by itself whether the drive has a 6821 PIA at $5000 (Multi-Speeder, DolphinDOS 3 cable) or whether the parallel cable is connected to the VIA1 (C64U with DolphinDos26UL), and transfers accordingly. Nothing needs to be set on the C64.

MS-NIBCOPY512 detects the firmware DolphinDos26UL by itself: when connecting, it reads the text of the power-on message “DOLPHINDOS26 C64U” from the drive's ROM. The following then applies:

- The status line shows `READY - DRIVE 08 (C64U)`.
- **Only the REU** is offered as storage (`STORAGE  : REU 512K`); the menu item STORAGE cannot be switched to MS-RAM.
- If **no REU** is switched on, STORAGE shows `NEED REU`. Reading, copying and writing are then refused with `C64U FIRMWARE: STORAGE NEEDS A REU` – switch on the REU in the C64U menu and restart MS-NIBCOPY512.
- The drive is automatically reset with `UJ` after every read and write operation: the payload uses the 8 KB drive RAM as a track buffer, and DolphinDos26UL then re-creates its working data – with the default settings (all switches R/F/V/P on, track count automatic); custom X settings such as `@XV-` must be set again afterwards. The status line shows `DRIVE: 00, OK,00,00`; after exiting, the drive can immediately be used normally again.

## 8.2 Minimum hardware requirements

| Component | Requirement |
|---|---|
| C64 | C64U, C64 or C128 in C64 mode |

| Component | Requirement |
|---|---|
| Storage (required) | REU – there is no MS-RAM with DolphinDos26UL; without an REU the program shows `NEED REU` and blocks reading and writing |
| Drive | DolphinDos26UL (or DolphinDos26 on the Multi-Speeder); not tested with DolphinDOS 2 |
| Connection | Parallel cable active (`P+` in the X status); without a cable the program reports `NO ANSWER FROM DRIVE CODE - CABLE?` |
| Drive RAM | 8 KB at $6000–$7FFF (track buffer) |
| REU | switch on in the C64U; 512 KB is enough for 35–40 tracks, for track 42 with half tracks 1 MB or more is recommended; if the space is not sufficient, `REU TOO SMALL FOR THE END TRACK` appears |
| Program | `MS-NIBCOPY512` on the disk `ms-tools.d64` |

## 8.3 Starting

```
LOAD"MS-NIBCOPY512",8
RUN
```

The programs on `ms-tools.d64` are packed with **Exomizer** and unpack themselves after `RUN` (MS-NIBCOPY512 therefore occupies 58 instead of 102 blocks). `LIST` still shows name, build number and author.




*Figure 6: Main menu of MS-NIBCOPY512 on the C64U: settings on the left, actions on the right; firmware detected ("READY - DRIVE 08 (C64U)"), storage fixed in the REU*

After starting, the program loads the saved presets (`MS-NIBCOPY.CFG`) and shows the main menu in **two columns**: on the left **SETTINGS**, on the right **ACTIONS**. Control is with the cursor keys: **CRSR up/down** goes through all items in order (from STORAGE on to COPY DISK), **CRSR left/right** changes a setting or jumps from the right column to the left, **RETURN** executes, ← leaves the menu or the program, **RUN/STOP** aborts a running operation.

> While MS-NIBCOPY512 is reading or writing, the drive's **red LED** is lit. Afterwards DolphinDos26UL takes over the LED again itself.

## 8.4 Main menu

| Menu item | Meaning |
|---|---|
| **SETTINGS** (left) | |
| PRESET | STANDARD (tracks 1–35, without half tracks), EXTENDED (up to track 42 with half tracks), CUSTOM 1–3 (freely adjustable) |
| END TRACK | last track: STANDARD 35–40, EXTENDED 36–42, CUSTOM 1–42. After IMPORT G64 FILE this shows the last track of the file; the value then also applies to the next read (permanently only with SAVE PRESETS) |
| HALFTRACKS | also read half tracks (empty and duplicate ones are discarded) |
| EDIT PRESET … | Submenu (see below) |
| DRIVE | Drive 8–11; if no device answers, `DRIVE nn: NOT PRESENT` appears |
| STORAGE | Storage: **REU** (size is detected, e.g. `REU 512K`) or `MS-RAM` (Multi-Speeder only). With DolphinDos26UL fixed to REU, or `NEED REU` without a REU |
| **ACTIONS** (right) | |
| COPY DISK | Read the source, then write the target disk |
| READ DISK | read only (into the REU) |
| WRITE DISK | Write the memory content to disk (read or imported tracks); then `ANOTHER COPY?` for further copies |
| IMPORT G64 FILE | Load a **G64 file** from a drive (e.g. SD2IEC) into memory, then write it to disk with WRITE DISK (chapter “Importing a G64 file”) |
| EXPORT G64 FILE | Write the read tracks as a **G64 file** to another drive, e.g. an SD2IEC (chapter “Exporting as a G64 file”) |
| TRACK INFO | Table per track: density, length, alignment, syncs, faulty sectors, read attempts (page by page with CRSR) |
| DIRECTORY | show the directory of the disk in the selected drive (CRSR up/down: line, CRSR left/right: page, RETURN or ←: back); tracks read or imported into storage are kept |
| QUIT | Exit the program |

### 8.4.1 EDIT PRESET

| Setting | Meaning |
|---|---|
| START TRACK / END TRACK | first and last track (1–42) |
| HALFTRACKS | half tracks on/off |
| ALIGNMENT | Track start when writing: AUTO, GAP (largest gap), SECTOR 0, LONGSYNC, BAD GCR, AUTO GAP, RAW – as with nibtools |
| DENSITY | AUTO (determine density per track) or fixed 0–3 |
| RETRIES | Read attempts 1–9 (default 5) |
| VERIFY | read back and compare each written track (default ON, up to 3 attempts) |
| RESET TO DEFAULTS | Restore defaults |
| SAVE PRESETS TO DISK | Save the settings as `MS-NIBCOPY.CFG` to the tool disk (`INSERT MS-TOOLS DISK`) |
| BACK | back to the main menu |

## 8.5 Copying a disk

1. In the main menu, check that **STORAGE** is set to **REU** (automatic with DolphinDos26UL).
2. Choose PRESET: STANDARD for normal disks, EXTENDED for disks with tracks above 35 or half tracks.
3. Select **COPY DISK** and press RETURN.
4. `INSERT SOURCE DISK`: insert the original disk, RETURN. The tracks are read; the grid shows the progress (OPEN open, NOW in progress, OK, SPECIAL special track such as killer or without sync, ERR error). At the end, `READ OK: TRACKS 01-35` is shown.
5. `INSERT TARGET DISK`: insert the target disk, RETURN. The program first measures the capacity of the target disk and then writes each track, with verify.
6. After `WRITE OK` the program asks `ANOTHER COPY?`: RETURN for another copy, ← to finish.




*Figure 7: After READ DISK: tracks 1–35 read via the VIA1 cable and stored in the REU, the drive is then ready again ("DRIVE: 00, OK,00,00")*

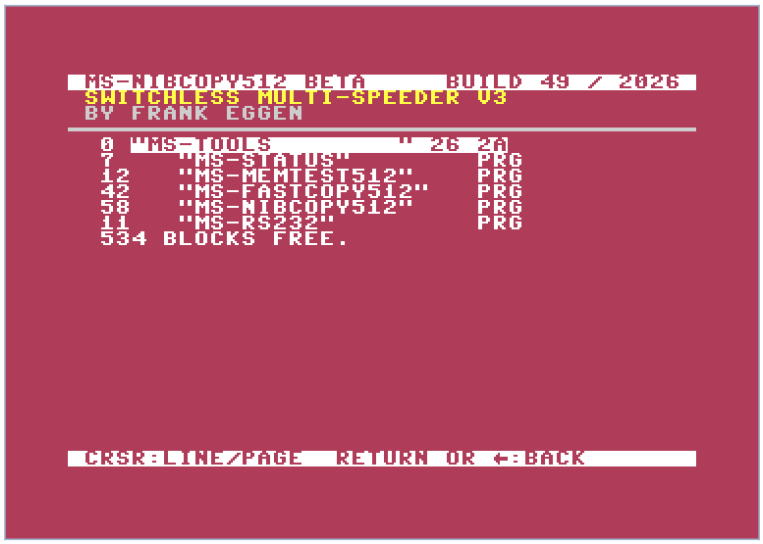



*Figure 8: MS-NIBCOPY512: directory shown with DIRECTORY*

| Message | Meaning |
|---|---|
| `TARGET DISK IS WRITE PROTECTED` | target disk is write-protected |

| Message | Meaning |
|---|---|
| `DRIVE SPEED NOT 280-320 RPM` | drive speed outside the permitted range |
| `VERIFY FAILED` | track not written correctly after three attempts |
| `DRIVE nn: NOT PRESENT` | no device answers at the selected drive |
| `NO DIRECTORY - SEE DRIVE MESSAGE` | DIRECTORY: no directory could be read, e.g. on an unformatted disk; the drive message is shown below |
| `NO NIBBLE DATA IN STORAGE` | nothing has been read yet |
| `REU TOO SMALL FOR THE END TRACK` | REU too small for the selected last track |
| `NO ANSWER FROM DRIVE CODE - CABLE?` | no connection via the parallel cable |
| `C64U FIRMWARE: STORAGE NEEDS A REU` | DolphinDos26UL detected, but no REU switched on |

## 8.6 Exporting as a G64 file

What MS-NIBCOPY512 has read can also be saved as a **G64 file** – the usual format for bit-exact disk images, used for example by VICE, the SD2IEC and the C64U itself. This turns an original disk into an image that can be archived or inserted directly on the C64U. The target is **another drive** with enough space, typically an **SD2IEC**: a G64 file is about 300 KB in size and does not fit on a 1541 disk.

1. First read the disk: **READ DISK** (or COPY DISK). Without read data the program reports `NO NIBBLE DATA IN STORAGE`.
2. Select **EXPORT G64 FILE** and press RETURN.
3. `EXPORT TO DRIVE: 09`: choose the target drive 8–11 with **CRSR left/right**, **RETURN** accepts. The program checks whether the device responds (otherwise `DRIVE 10: NOT PRESENT`). The first drive that is not the reading drive is suggested, afterwards the last one used.
4. `FILE NAME: NIBCOPY.G64`: type the name (up to 12 characters: A–Z, 0–9, space, `-`, `.`, `+`; **DEL** deletes). The extension `.G64` is appended; an existing file of the same name is replaced. **RETURN** starts the export, ← aborts.
5. The track grid shows the progress (`EXPORTING TRACK nn`). At the end, `EXPORT OK: NAME.G64` and the message of the target drive are shown, e.g. `DRIVE 09: 00, OK,00,00`.




*Figure 9: EXPORT G64 FILE: choose the target drive (here drive 9, e.g. an SD2IEC)*




*Figure 10: Enter the file name (.G64 is appended)*




*Figure 11: Export finished: file MEINSPIEL.G64 written to drive 9*

- Each track is aligned as when writing to disk and built with complete syncs; its length corresponds to the capacity of a track of this density at 300 rpm (longer tracks keep their

length). Tracks 36–42, half tracks, killer tracks and **weak bits** (spots without flux transitions) are retained.

- With DolphinDos26UL the data comes from the REU; the target may be any other device. On the Multi-Speeder with storage in the MS-RAM, the reading drive supplies the data and therefore cannot itself be the target (`MS-RAM USES THIS DRIVE - CHOOSE ANOTHER`).
- The duration depends on the target: a few minutes over the normal serial bus, considerably less with a JiffyDOS-capable target (e.g. SD2IEC) – the C64 kernal DolphinDos26 selects JiffyDOS automatically.

| Message | Meaning |
|---|---|
| `EXPORT OK: NAME.G64` | file written completely |
| `DRIVE nn: NOT PRESENT` | no device responds on the selected drive |
| `MS-RAM USES THIS DRIVE - CHOOSE ANOTHER` | Multi-Speeder with MS-RAM storage only: reading drive chosen as the target |
| `EXPORT FAILED: …` | Abort with RUN/STOP or transfer error; the message of the target drive appears below it (e.g. `72,DISK FULL`) |

## 8.7 Importing a G64 file

The reverse route: a **G64 file** – e.g. an image from the internet, from VICE, from nibtools or one exported earlier – is loaded from a drive into memory and then written to a real disk with **WRITE DISK**. The source is typically an **SD2IEC**.

1. Select **IMPORT G64 FILE** and press RETURN.
2. `IMPORT FROM DRIVE: 09`: with **CRSR left/right** choose the drive 8–11 on which the file is located, **RETURN** accepts (a missing device reports `DRIVE 10: NOT PRESENT`).
3. `FILE NAME:`: type the name without `.G64` (up to 12 characters) and press RETURN. The last used name is suggested, the first time NIBCOPY.
4. The file is read track by track (`IMPORTING TRACK nn`), the grid shows the progress. At the end, `IMPORT OK: TRACKS 01-35` is shown (or up to the last track of the file); END TRACK in the main menu then shows this last track.
5. Insert the target disk, select **WRITE DISK** and write as when copying (with verify, then `ANOTHER COPY?`).




*Figure 12: IMPORT G64 FILE: choose the drive with the G64 file (here drive 9)*




*Figure 13: Import finished: tracks 1–35 from MEINSPIEL.G64 in the REU, ready for WRITE DISK*

- Each track of the file is evaluated like a read track (length, alignment, sectors); the bytes are taken over unchanged when writing. In this way short syncs, killer tracks, tracks without sync, half tracks and weak bits are also retained.
- The tracks in the file must be in ascending order, as with VICE, nibtools and EXPORT G64 FILE; otherwise `IMPORT FAILED: TRACKS NOT IN ORDER` appears. Files with changing density within a track are taken over with the standard density of the zone.
- With DolphinDos26UL, loading is into the REU; the source may be any drive, including drive 8 with a disk on which the file is located. On the Multi-Speeder with storage in the MS-RAM, the reading drive cannot at the same time be the source.

| Message | Meaning |
|---|---|
| `IMPORT OK: TRACKS 01-35` | file loaded, ready for WRITE DISK |
| `IMPORT FAILED: NOT A VALID G64 FILE` | not a G64 file, file too short or faulty |
| `IMPORT FAILED: TRACKS NOT IN ORDER` | tracks of the file not in ascending order |

| Message | Meaning |
|---|---|
| `DRIVE 09: 62,FILE NOT FOUND,00,00` | file not found on the drive (message from the drive) |

## 8.8 Why slower than nibtools on a PC?

MS-NIBCOPY512 follows the same method as **nibtools**, but takes noticeably longer than nibtools on a PC with an XUM1541 (ZoomFloppy). This is not a fault; it comes mainly from where the computing happens. One full track when reading, measured in the emulator: The values come from the Multi-Speeder setup with the parallel cable on the PIA; on the C64U the transfer in particular may differ.

| Step | Time | Who does the work |
|---|---|---|
| Read track | 220 ms | drive |
| Transfer 8 KB to the C64 | 363 ms | drive and C64, over the parallel cable |
| Count syncs | 126 ms | C64 |
| Determine track start, length and alignment | 335 ms | C64 |
| Check sectors | 389 ms | C64 |

- **Computing power:** More than half of the time goes into the analysis. nibtools does the same steps (density, track length, alignment, sectors, invalid GCR) on the PC, where they hardly matter. Here the 8-bit processor of the C64 does the work at about 1 MHz.
- **Drive:** This part makes no big difference. nibtools also reads with the same mechanism and, on a 1541, fetches the raw data over a parallel cable, 8 KB per track.
- **Memory:** The C64 has 64 KB of RAM. The tracks therefore sit in the REU; they are analysed in the RAM of the C64.
- **Checking when writing:** Here VERIFY by default reads each track back and compares it, with up to 3 attempts. In nibtools this check is off by default. VERIFY OFF (EDIT PRESET) makes writing faster, but write errors then go unnoticed.
- **Difficult tracks:** Tracks with invalid GCR take several reads on both sides. On the C64 the comparison comes on top.

## 8.9 Test status

- DIRECTORY, END TRACK after an import, the file name suggestion with arbitrary RAM contents and the message for a missing drive have been checked; reading and writing with this build also with DolphinDos26UL.
- With DolphinDos26UL, parallel cable on VIA1 (without PIA, as on the C64U), 8 KB RAM and 512 KB REU: reading (tracks 1–35) into the REU and writing with verify successful; the directory of the copy matches the original. Detection of the C64U firmware (REU only, blocked without REU) and the automatic reset of the drive have also been checked.
- Copy protection (Multi-Speeder, G64 test images): 10 of 11 copies start; this includes the V-MAX protection with very short syncs on track 20. Weak bits are detected (a spot counts as a weak bit only when three reads report it; up to ten additional reads) and written as “no flux transition”. In the emulator, weak bits read only weakly at random, so with one test image about every fourth copy does not start there – in that case repeat the copy.

- G64 export (drive 9 as a directory like an SD2IEC): test image exported from the REU and from the Multi-Speeder RAM, the file matches the source track by track and in the densities; the exported G64 starts.
- G64 import: test image imported from the REU and from the Multi-Speeder RAM and written – WRITE OK with verify, the disk matches the file; the copy written from the imported G64 starts. The red LED is lit while reading and is off afterwards.
- Spots with four and five consecutive zero bits are read falsified by every 1541 (after three zero bits its bit counter itself generates a 1). On tracks with density 3, MS-NIBCOPY512 therefore reads them once more with density 0 and restores them bit-exactly (one track of a test image; about 2 seconds per affected track). With long gaps, the seam of the copy is now at the start of the gap instead of directly before its end.
- Not yet copyable: a test image whose loader also checks such spots on tracks 31–38; these have density 0 and cannot be read more slowly.

> **Legal note:** Copies may only be made of disks to which one holds the rights (e.g. backup copies of one's own originals).

# 9 MS-FASTCOPY512

MS-FASTCOPY512 copies disks with normal format (data copy, no copy protection) track by track and shows the progress in the track grid. With DolphinDos26UL the **REU** serves as memory; the Multi-Speeder RAM and the MS-Addon-API do not exist here. For copy-protected disks, MS-NIBCOPY512 remains the tool to use.

## 9.1 Minimum hardware requirements

| Component | Minimum requirement |
|---|---|
| C64 | C64U, C64 or C128 in C64 mode |
| Storage (required) | REU – there is no MS-RAM with DolphinDos26UL; without an REU nothing is copied |
| Drive | DolphinDos26UL (C64U or real DolphinDOS 2 hardware) |
| Connection and kernal | active parallel cable on VIA1 (emulated on the C64U) and a DolphinDOS kernal on the C64, `C64DolphinDos26.bin` recommended; without a parallel connection the REU is not offered |
| REU | one slot (`REU 1`) from 256 KB, two slots from 512 KB; switch it on in the C64U menu |
| Otherwise | without a usable REU the program only shows the directory (DIRECTORY); copying is not possible |

## 9.2 Operation

```
LOAD"MS-FASTCOPY512",8
RUN
```

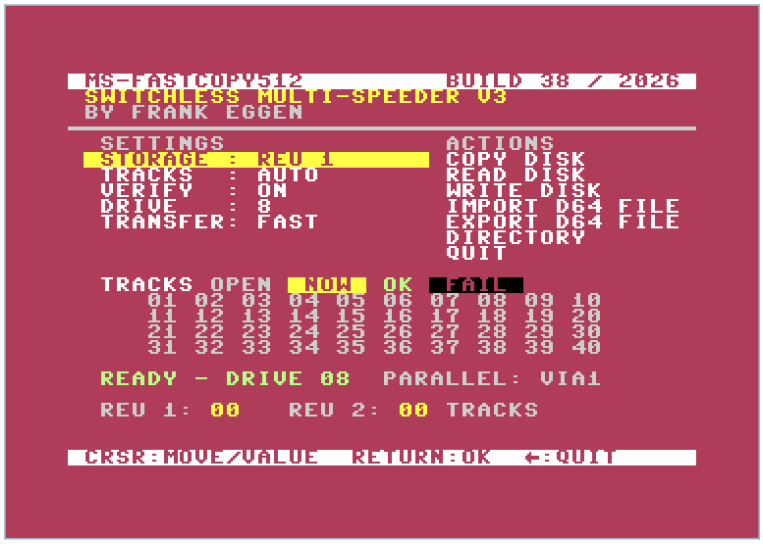



*Figure 14: MS-FASTCOPY512 on the C64U: parallel cable on VIA1 detected, memory REU 1*

| Item | Meaning |
|---|---|
| STORAGE | `REU 1`/`REU 2`, 256 KB each at the top end of the REU; one slot from 256 KB of REU, two from 512 KB |

| Item | Meaning |
|---|---|
| TRACKS | 35, 40 or AUTO (checks track 36) |
| VERIFY | verify after writing |
| TRANSFER | `FAST` (default): own drive code, fast, screen off while copying; `DOS`: the previous, slower path, for problems |
| DRIVE | drive 8–11; on change the program checks the parallel connection |
| COPY DISK | read the source, then write to the target disk; `ANOTHER COPY?` allows further copies |
| READ DISK / WRITE DISK | read only / write only; the program can format an unformatted target |
| IMPORT D64 FILE | load a **D64 file** from a drive (e.g. SD2IEC) into the REU, then write it with WRITE DISK |
| EXPORT D64 FILE | write the slot contents as a **D64 file** to a drive |
| DIRECTORY | directory of the selected drive |

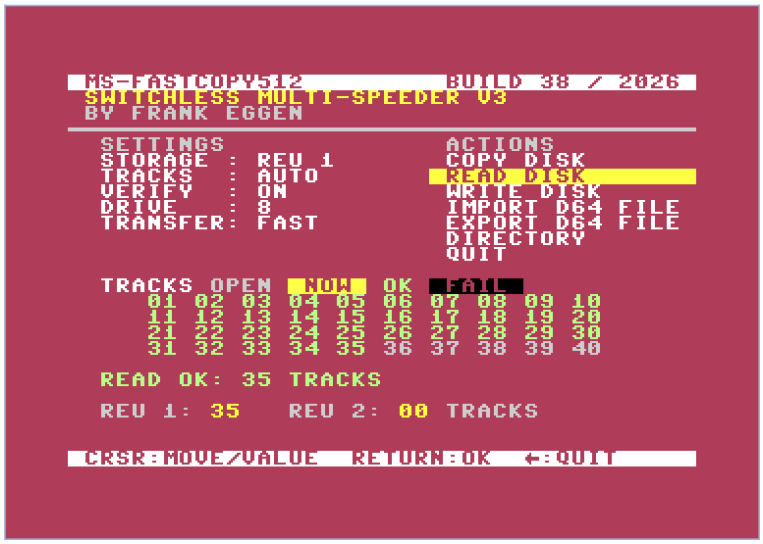



*Figure 15: Disk read into REU 1, ready for WRITE DISK or EXPORT D64 FILE*

- The screen shows `READY - DRIVE 08  PARALLEL: VIA1`. Without a parallel connection (`NO PARALLEL`, also with `@XP-` or a kernal without DolphinDOS) or without a large enough REU (`NO REU`) the program refuses to copy.
- In the emulator this takes about 30 s reading and 44 s writing with verify for 35 tracks with `TRANSFER: FAST`, and about 90 s and 130 s with `TRANSFER: DOS`.
- The data stays as long as the REU has power. MS-NIBCOPY512 uses the same REU from the bottom, MS-FASTCOPY512 from the top; with a small REU the data can overwrite each other, which is not always detected.
- D64 import and export: dialogs like those for G64 files (`EXPORT TO DRIVE: 09`, `FILE NAME:`). 35 and 40 tracks are accepted, also with an error table (it is ignored); other sizes report `IMPORT FAILED: NOT A VALID D64 FILE`.

## 9.3 Speed

Measured on real hardware (C64U with a 1541 with Multi-Speeder V3, parallel cable on VIA1, kernal `C64DolphinDos26.bin`), 35 tracks, REU storage, to the nearest second:

| Firmware | Reading | Writing with verify | Writing without verify |
|---|---|---|---|
| DolphinDOS 2.0 (original, 8 KB RAM) | 82 s | 121 s | 114 s |
| DolphinDos26DD2 (Multi-Speeder) | 82 s | 120 s | 113 s |

With DolphinDos26UL, the VICE emulator gives the same values (83 s, 120 s, 117 s). With an REU every sector travels over the parallel cable to the C64 and back; the drive RAM does not serve as storage.

# 10 MS-15SECCOPY

MS-15SECCOPY is a new version of **15 Seconds Copy**, written by **Mike Pall** in 1986. It copies disks with normal format (no copy protection) as raw GCR over the parallel cable and, like the original, shows every track and sector in a grid. The interface follows MS-FASTCOPY512; new is the choice between a cable on VIA1 and on the PIA. The program is on `ms-tools.d64` and also on a separate disk (`ms-15seccopy.d64`).

> **Warning:** MS-15SECCOPY uses the REU as a buffer. Data stored there, including slots of MS-FASTCOPY512 and MS-NIBCOPY512, may be overwritten.

## 10.1 Minimum hardware requirements

| Component | Minimum requirement |
|---|---|
| C64 | C64 or C128 in C64 mode, also C64U |
| REU (required) | at least 256 KB for 35 tracks, 512 KB for 40 tracks; without an REU, `NEEDS AN REU WITH AT LEAST 256 KB` appears |
| Parallel cable (required) | on VIA1 of the 1541 (SpeedDOS cable) or on the PIA (DolphinDOS 3, Multi-Speeder). The program proposes the matching cable, and `CABLE` changes it. If there is no PIA at `$5000` (for example under DolphinDOS 2.0 on the Multi-Speeder), `PIA` is locked and `NO PIA AT $5000 IN THIS DRIVE ROM` appears. It does not check whether a cable is actually connected |
| Drive | 1541 as device 8. The original ROM is enough; DolphinDos26, Multi-Speeder and RAM expansion are **not** needed |
| C64 kernal | the original kernal is enough; a DolphinDOS kernal is **not** needed |

## 10.2 Operation

```
LOAD"MS-15SECCOPY",8
RUN
```

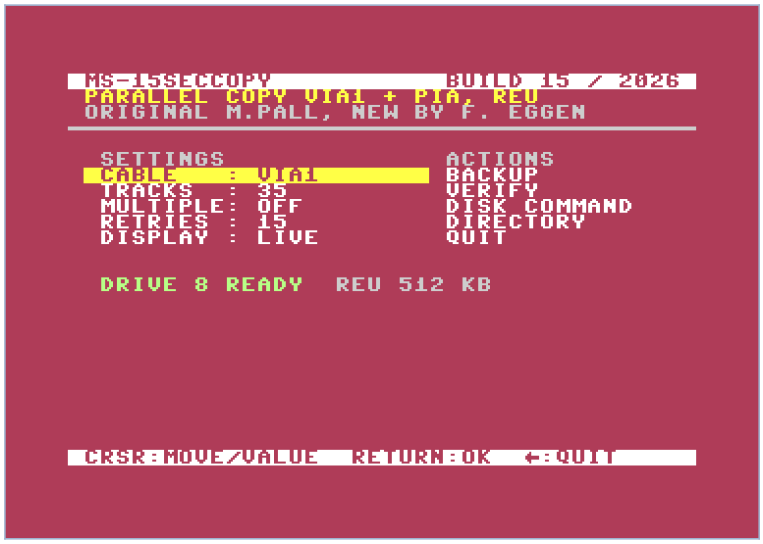



*Figure 16: MS-15SECCOPY: menu with detected parallel cable and REU*

| Item | Meaning |
|---|---|
| CABLE: VIA1/PIA | cable used for copying |
| TRACKS: 35/40 | tracks; 40 tracks need an REU with 512 KB |
| MULTIPLE: OFF/ON | several copies of the same source |
| RETRIES: 5–15 | read attempts per sector |
| DISPLAY: LIVE/BLNK | `LIVE`: the grid appears briefly between tracks; `BLNK`: as in the original, the screen stays dark while copying |
| BACKUP | copy a disk; the program asks for source and target |
| VERIFY | check a disk, result in the grid |
| DISK COMMAND | send a command to the drive and show the message |
| DIRECTORY | show the directory |
| QUIT | back to BASIC |

On real hardware (C64U, 1541, REU), BACKUP and VERIFY ran over all 683 blocks without errors, on VIA1 under DolphinDOS 2.0 and on the PIA under DolphinDos26DD2; 35 tracks took at most 12 s reading and 18 s writing. Source and target were the same disk.

# 11 Acknowledgements

DolphinDos26UL stands on the shoulders of great predecessors. My thanks go to:

| Project | Developer | Contribution |
|---|---|---|
| **DolphinDOS 3.0** | Günther Jilg (software, DolphinDOS and DolphinCopy, 1985–1987) and Jan Bubela (hardware, 1985–1988), Dolphin Software, Oberursel | Basis of the drive ROM and the C64 kernal: parallel transfer, track cache, X commands, 40 tracks, editor and keyboard functions; the parallel routines for the VIA1 cable come from **DolphinDOS 2.0** |
| **JiffyDOS** | Mark Fellows; distributed by Creative Micro Designs (CMD), now Retro Innovations | The JiffyDOS transfer protocol that the kernal uses for JiffyDOS devices and SD2IEC |
| **nibtools** | Pete Rittwage, based on MNIB by Markus Brenner | Methods and algorithms of MS-NIBCOPY512 (reading, density detection, alignment, writing) |
| **15 Seconds Copy** | Mike Pall, 1986 | pioneering work on fast parallel copying; basis of MS-15SECCOPY |
| **Exomizer** | Magnus Lind | Packer with which the programs on `ms-tools.d64` are packed |
| **SD2IEC** | Ingo Korb and contributors | SD card drive whose commands the SD2IEC function keys use |
| **VICE** | the VICE team | Emulator for development and tests |

**Special thanks go to Jim Brain of RETRO Innovations** for his support with the JiffyDOS implementation of the C64/C128 kernal DolphinDos26.

**My thanks also go to Mike Pall** for his pioneering work with the original 15 Seconds Copy in 1986.

> **Please check: JiffyDOS licence.** The kernal contains JiffyDOS transfer routines that were taken identically, instruction for instruction, from a JiffyDOS kernal; the JiffyDOS firmware for the drive is not included. Jim Brain describes it as follows: for code that implements the same transfer behaviour but is not an exact copy of the JiffyDOS routines, no licence is needed. If the transfer code is the same, in his assessment a licence would be required on the part of the end user. Every user is therefore asked to check for themselves whether they should acquire an official JiffyDOS licence because of their use. Further information: www.jiffydos.com

Thanks also to everyone who has preserved the original DolphinDOS manuals as text, and to the Commodore community, which keeps the C64 alive to this day.

# 12 About the developer

DolphinDos26UL, the C64 kernal DolphinDos26, the firmware DolphinDos26 for the 1541 Switchless 8x Multi-Floppy-Speeder V3 and the MS-Tools were developed by **Frank Eggen**. All source code is fully commented (ca65 assembler).

| Topic | Address |
|---|---|
| Contact | frankeggen@emden.net |
| Projects on GitHub | https://github.com/fraegg |

On GitHub you can find further projects around the Commodore 64 and the 1541. Questions, bug reports and experience reports – especially from the real C64U – are welcome.

# 13 Disclaimer

DolphinDos26UL, the C64 kernal DolphinDos26 and the MS-Tools are a **private hobby project**. They are provided without any warranty, as is. No guarantee is given for function, freedom from errors or fitness for a particular purpose.

- Use is **at your own risk**. No liability is accepted for damage to hardware, disks or data, or for any other consequences.
- Before use, backup copies of important disks should be made. Formatting, writing and nibble copying change disks irrevocably.
- The C64 kernal also runs on real hardware; MS-NIBCOPY512 has so far only been tested on real hardware with the Multi-Speeder and normal disks.
- Copyright must be observed when copying disks.

# 14 Support through a donation

Development takes a lot of time. Anyone who would like to support the project can voluntarily send the developer a donation via PayPal:

> **PayPal:** Donation to Frank Eggen
>
> Link to type in: https://www.paypal.com/donate/?cmd=_s-xclick&hosted_button_id=Q8HXKYARXKT4L&ssrt=1714757590172

Many thanks for any support!

# 15 Copyright and terms of use

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